

Part A: Negative Number Operations (Complex)*Evaluate each expression completely.*

1. $-27 + 10 \cdot 5$

9. $24 - 4 \cdot (2 + 24)$

2. $132 \div (-12) - (-20)$

10. $4^3 - (-19) \cdot 11$

3. $-32 + 7 \cdot (-13)$

11. $(-6)^2 - (-26) \cdot 15$

4. $-17 + (-15) \cdot 9$

12. $-10 - 8 \cdot ((-7) + (-10))$

5. $54 \div 6 - (-39)$

13. $-8 - 14 \cdot ((-2) + (-8))$

6. $-70 \div 5 - 12$

14. $38 - 5 \cdot ((-4) + 38)$

7. $40 + 12 \cdot 8$

15. $(5 + 28)^2 \div 9$

8. $-35 + 4 \cdot (-4)$

16. $3^2 - (-7) \cdot 8$

Part B: Fractions*Evaluate. Write every answer in lowest terms.*

1. $\frac{1}{3} + \frac{1}{5}$

4. $\frac{1}{4} - \frac{2}{5}$

2. $2 - \frac{1}{3}$

5. $\frac{4}{9} \div \frac{2}{3}$

3. $3 + \frac{1}{10}$

6. $\frac{7}{8} \div \frac{8}{9}$

7. $5\frac{1}{2} \cdot \frac{1}{3}$

12. $5\frac{1}{2} - 7\frac{1}{2}$

8. $\frac{1}{4} \cdot \frac{4}{5}$

13. $\frac{35}{50}$

9. $5\frac{3}{4} + 3\frac{1}{3}$

14. $\frac{33}{36}$

10. $1\frac{9}{10} + 6\frac{1}{2}$

15. $\frac{63}{14}$

11. $1\frac{3}{5} + 4\frac{4}{5}$

16. $\frac{7}{84}$

Part C: Exponent Properties

Simplify. Write answers with positive exponents.

1. $x^4 \cdot 4x^2$

7. $(5x^7)^3$

2. $2x^6 \cdot x^6$

8. $(6x^7)^2$

3. $2x^6 \cdot 4x^2$

9. $(6x^8)^2$

4. $\frac{2x^7}{x^2}$

10. $2 \cdot 5^{-3}$

5. $\frac{x^{11}}{x^6}$

11. $2 \cdot (5^1)^{-1}$

12. $7^4 \cdot 7^{-3}$

6. $\frac{4x^7}{x^6}$

13. $5^2 \cdot 5^{-3}$

14. $(4x^{-1})^3 \cdot x^{11}$

16. $(2x^{-1})^3 \cdot x^4$

15. $(6x^{-3})^2 \cdot x^{16}$

Part D: Simplifying and Factoring Expressions*Simplify each expression completely.*

1. $10x - 3 + 7x + 8$

9. $(-2x - 10) - (6x + 4)$

2. $10x - 11 + x + 2$

10. $(10x - 12) + (-4x - 4)$

3. $12x - 4 + 2x + 8$

11. $(x - 2) + (4x + 6)$

4. $3x + 10 + 10x - 1$

12. $(10x + 4) + (10x + 12)$

5. $4(-11x - 11) + 5x + 12$

13. $-49x + 7$

6. $12(8x + 4) + 5x + 4$

14. $-36x + 42$

7. $2(-2x + 8) + 5x + 10$

15. $32x + 8$

8. $-11(5x + 2) - 5x + 9$

16. $40x - 25$

Part E: Solving for x*Solve each equation for x. Show your steps.*

1. $8x = 32$

3. $2x = 6$

2. $6x = -24$

4. $8x + 9 = -15$

11. $8x - 6 = 6x - 2$

5. $9x + 5 = 14$

12. $3x - 8 = -5x + 72$

6. $5x + 7 = 32$

13. $-\frac{2}{3}x - 10 = -\frac{40}{3}$

7. $4(2x - 3) + 4 = 16$

14. $\frac{x - 1}{2} + 2 = \frac{13}{2}$

8. $7(2x + 12) - 5 = 205$

15. $\frac{8x + 74}{2} = \frac{20}{4}$

9. $6(4x - 6) - 8 = 172$

16. $\frac{8x - 3}{12} = \frac{9x - 39}{4}$

10. $7x - 10 = 12x + 10$

Part F: Solving Inequalities

Solve each inequality for x .

1. $x - 2 < 3$

4. $x + 9 \geq -9$

2. $-3x \leq -24$

5. $-2x - 3 \geq -9$

3. $8x > 40$

6. $10x + 3 \geq 13$

12. $-4x - 6 \leq -2x + 16$

7. $-8x - 12 \leq 20$

13. $\frac{x}{3} - 7 \geq -\frac{26}{3}$

8. $-12x + 11 \geq 35$

14. $-3x + 8 \geq 29$

9. $-8x + 5 \leq 7x - 25$

15. $\frac{x}{2} + 5 < \frac{25}{2}$

10. $6x - 12 < 9x - 30$

16. $\frac{x - 14}{2} + 15 < \frac{25}{2}$

11. $x + 2 < -8x + 74$

Part G: Solving Absolute Value Equations

Solve each equation for x .

1. $|x - 4| = 3$

5. $|5x + 40| = 45$

2. $|x| = 8$

6. $|5x + 30| = 50$

3. $|x - 2| = 8$

7. $|5x + 25| = 65$

4. $|x + 8| = 3$

8. $|3x - 24| = 9$

9. $\frac{|x-1|}{4} + 15 = 17$

11. $5|x+7| + 10 = 40$

10. $4|x-14| + 11 = 19$

12. $4|x-12| + 3 = 27$

Part H: Proportions

Solve each proportion for x .

1. $\frac{8}{10} = \frac{24}{x}$

4. $\frac{7}{13} = \frac{x}{39}$

2. $\frac{11}{13} = \frac{55}{x}$

5. $\frac{11}{11} = \frac{44}{x}$

3. $\frac{1}{x} = \frac{4}{12}$

6. $\frac{9}{10} = \frac{36}{x}$

Part I: Unit Rate and Price Comparison

Find the unit rate, or determine which option is the better price per unit.

1. Ethan pays \$152 for 8 pencils. Find the price per pencil.

2. Zoe pays \$112 for 7 erasers. Find the price per eraser.

3. Liam pays \$36 for 4 erasers. Find the price per eraser.

4. Mia pays \$55 for 11 notebooks. Find the price per notebook.

5. Zoe pays \$80 for 10 pencils. Find the price per pencil.

6. Option A: \$240 for 12 granola bars. Option B: \$126 for 7 granola bars. Which option is the better price per granola bar?

7. Option A: \$48 for 6 notebooks. Option B: \$88 for 8 notebooks. Which option is the better price per notebook?

8. Option A: \$121 for 11 erasers. Option B: \$10 for 5 erasers. Which option is the better price per eraser?

9. Option A: \$95 for 5 granola bars. Option B: \$110 for 10 granola bars. Which option is the better price per granola bar?

10. Option A: \$14 for 7 juice boxes. Option B: \$200 for 10 juice boxes. Which option is the better price per juice boxe?

Answer Key

Part A: Negative Number Operations (Complex)

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|-----------|----------|------------|
| 1. 23 | 7. 136 | 13. 132 |
| 2. 9 | 8. -51 | 14. -132 |
| 3. -123 | 9. -80 | 15. 121 |
| 4. -152 | 10. 273 | 16. 65 |
| 5. 48 | 11. 426 | |
| 6. -26 | 12. 126 | |

Part B: Fractions

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|--------------------|--------------------|---------------------|
| 1. $\frac{8}{15}$ | 7. $1\frac{5}{6}$ | 13. $\frac{7}{10}$ |
| 2. $1\frac{2}{3}$ | 8. $\frac{1}{5}$ | 14. $\frac{11}{12}$ |
| 3. $3\frac{1}{10}$ | 9. $9\frac{1}{12}$ | 15. $4\frac{1}{2}$ |
| 4. $-\frac{3}{20}$ | 10. $8\frac{2}{3}$ | 16. $\frac{1}{12}$ |
| 5. $\frac{2}{3}$ | 11. $6\frac{3}{5}$ | |
| 6. $\frac{63}{64}$ | 12. -2 | |

Part C: Exponent Properties

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|--------------|---------------------|-------------------|
| 1. $4x^6$ | 7. $125x^{21}$ | 13. $\frac{1}{5}$ |
| 2. $2x^{12}$ | 8. $36x^{14}$ | 14. $64x^8$ |
| 3. $8x^8$ | 9. $36x^{16}$ | 15. $36x^{10}$ |
| 4. $2x^5$ | 10. $\frac{2}{125}$ | 16. $8x$ |
| 5. x^5 | 11. $\frac{2}{5}$ | |
| 6. $4x$ | 12. 7 | |

Part D: Simplifying and Factoring Expressions

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|----------------|----------------|------------------|
| 1. $17x + 5$ | 7. $x + 26$ | 13. $7(-7x + 1)$ |
| 2. $11x - 9$ | 8. $-60x - 13$ | 14. $6(-6x + 7)$ |
| 3. $14x + 4$ | 9. $-8x - 14$ | 15. $8(4x + 1)$ |
| 4. $13x + 9$ | 10. $6x - 16$ | 16. $5(8x - 5)$ |
| 5. $-39x - 32$ | 11. $5x + 4$ | |
| 6. $101x + 52$ | 12. $20x + 16$ | |

Part E: Solving for x

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|-------------|--------------|--------------|
| 1. $x = 4$ | 7. $x = 3$ | 13. $x = 5$ |
| 2. $x = -4$ | 8. $x = 9$ | 14. $x = 10$ |
| 3. $x = 3$ | 9. $x = 9$ | 15. $x = -8$ |
| 4. $x = -3$ | 10. $x = -4$ | 16. $x = 6$ |
| 5. $x = 1$ | 11. $x = 2$ | |
| 6. $x = 5$ | 12. $x = 10$ | |

Part F: Solving Inequalities

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|-----------------|------------------|-----------------|
| 1. $x < 5$ | 7. $x \geq -4$ | 13. $x \geq -5$ |
| 2. $x \geq 8$ | 8. $x \leq -2$ | 14. $x \leq -7$ |
| 3. $x > 5$ | 9. $x \geq 2$ | 15. $x < 15$ |
| 4. $x \geq -18$ | 10. $x > 6$ | 16. $x < 9$ |
| 5. $x \leq 3$ | 11. $x < 8$ | |
| 6. $x \geq 1$ | 12. $x \geq -11$ | |

Part G: Solving Absolute Value Equations

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|--------------------------|-------------------------|---------------------------|
| 1. $x = 1$ or $x = 7$ | 5. $x = -17$ or $x = 1$ | 9. $x = -7$ or $x = 9$ |
| 2. $x = -8$ or $x = 8$ | 6. $x = -16$ or $x = 4$ | 10. $x = 12$ or $x = 16$ |
| 3. $x = -6$ or $x = 10$ | 7. $x = -18$ or $x = 8$ | 11. $x = -13$ or $x = -1$ |
| 4. $x = -11$ or $x = -5$ | 8. $x = 5$ or $x = 11$ | 12. $x = 6$ or $x = 18$ |

Part H: Proportions

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|-------------|-------------|-------------|
| 1. $x = 30$ | 3. $x = 3$ | 5. $x = 44$ |
| 2. $x = 65$ | 4. $x = 21$ | 6. $x = 40$ |

Part I: Unit Rate and Price Comparison

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|---------------------|-------------------|--------------|
| 1. \$19 per pencil | 5. \$8 per pencil | 9. Option B |
| 2. \$16 per eraser | 6. Option B | 10. Option A |
| 3. \$9 per eraser | 7. Option A | |
| 4. \$5 per notebook | 8. Option B | |